CSC510-05 Analysis of Algorithms I

Department:	Computer Science	**Semester:**	Spring 25
Course:	CSC510	**Section:**	05
Modality:	In-person Instruction	**Units:**	3
Instructor:	Jose Ortiz	**Email:**	jortizco@sfsu.edu
Lecture Time:	MoWe 02:00PM - 3:15PM	**Room:**	Thornton Hall 409

Course Website: All the course materials will be available through Canvas. https://canvas.sfsu.edu/

Office Hours: Office hours are held online on Tuesdays from 9:00 AM to 10:00 AM and Thursdays from 1:30 PM to 2:30 PM. No appointment is required, but you must email in advance. All relevant details and links are posted on Canvas.

Prerequisites:

- A grade of C or better in CSC 340 (Programming Methodology)
- A grade of C or better in Math 324 (Probability and Statistics in Computing).
- All the course's prerequisites are strictly enforced, and concurrent enrollment is not allowed.

Recommended Reading: This course doesn't use a book because all the class' material will be accessible from Canvas. However, in case students need a good reference, I recommend the book: "Foundations of Algorithms by Neapolitan."

Description: Detailed study of the basic notions of the design of algorithms and the underlying data structures. Several measures of complexity are introduced. Emphasis is on the structure, complexity, and efficiency of algorithms. layers.

Outline of the Course:

- Algorithms: concepts and complexity analysis from pseudo-code
- Complexity, Time Complexity, Space Complexity and Running Time.
- Notions of complexity order: definitions, applications and implications
- The sorting problem: common sorting algorithms, complexity analysis, and comparison
- The divide-and-conquer methodology: application in the sorting and search problems' domain
- Recurrence relations: definition, application and approaches to solve them
- Dynamic programming: concepts, main implementation procedure, and applications, including string matching and parsing algorithms
- The All-pairs Shortest Paths problem: Floyd-Warshall's dynamic programming solution
- The Traveling Salesperson Problem: optimal structure analysis and a dynamic programming solution
- The LCS (Longest Common Subsequence) Problem: optimal structure analysis and a dynamic programming solution
- The greedy approach: concepts, main implementation steps, and applications

- Randomized Algorithms
- The Minimum Spanning Tree problem: Prim's and Kruskal's greedy algorithms
- The Single Source Shortest Paths problem: Dijkstra's greedy algorithm
- The Scheduling problem: different variants and their corresponding greedy algorithms
- The 0-1 Knapsack problem: a greedy algorithm
- Comparison of the dynamic programming and greedy approaches
- The backtracking optimization technique: concepts and applications
- The n-Queen problem: a solution with backtracking optimization
- The Sum-of-Subsets problem: a solution with backtracking optimization
- Branch and Bound algorithms: concepts, main implementation procedure, and applications
- The NP = P problem

Grading Policy Overview: To encourage authentic learning and reduce incentives for improper use of AI tools, the course grading is structured as follows:

1. **Three Midterm Exams (45%):**
   - Each midterm contributes 15% to your final grade.
   - One of your lowest (non-zero) midterm grades will be replaced by your highest midterm grade. This policy provides an opportunity to recover from one poor performance.
2. **Final Exam (30%):**
   - A comprehensive, in-person, proctored exam covering all course material.
   - Notes, books, or unauthorized electronic aids are not permitted.
3. **Group or Individual Project (15%):**
   - You can work solo or team up with up to 2 other classmates (maximum of 3 students per group).
   - Collaborate to solve a real-world algorithmic problem using a step-by-step approach.
   - Actively engage in group discussions to analyze and refine the problem-solving process.
4. **Participation (10%):**
   - Includes random attendance checks or in-class activities.
5. **In-Class Extra Credit:**
   - Extra credit opportunities are available exclusively to students who attend lectures.
6. **Practice Exercises:**
   - Practice exercises are for practice only and will not be collected or graded.
   - Completing these exercises is highly recommended to prepare for midterms and the final exam.
   - Students who regularly practice tend to perform significantly better on exams.

Important Dates:

- Last Day to Drop/Withdraw Classes Without a W: Monday February 17 2025
- Grading Option Deadline (This includes Cr/No CR, Audit, or Letter Grade): Friday May 09 2025
- Midterm Exams: See tentative schedule below
- Final Exam: Monday May 19, 2:45PM - 4:45PM

Grades: The final letter grade for the course will be computed using the following distribution. I reserve the right to grade more leniently than this, but I will not grade more strictly.

Highest	Lowest	Letter Grade
100	93	A
92.99	90	A-
89.99	87	B+
86.99	83	B
82.99	80	B-
79.99	77	C+
76.99	73	C
72.99	70	C-
69.99	67	D+
66.99	63	D
62.99	60	D-
59.99	0	F

Tentative Schedule:

Week	Monday	Wednesday	Midterm Exams Dates
1	Intro to the Course	Complexity of Algorithms I	
2	Time Complexity, Space Complexity	Running Time	
3	Asymptotic Order, Limit Ratio	Recurrence Relations I	
4	Recurrence Relations II	Master Theorem	
5	Introduction to Pseudocode	Midterm #1	Wednesday February 26
6	Divide and Conquer I	Divide and Conquer II	
7	Backtracking Algorithms I	Backtracking Algorithms II	
8	Branch and Bound I	Branch and Bound II	
9	Spring Break	Spring Break	
10	Cesar Chavez Observed (No Class)	Midterm #2	Wednesday April 02
11	DP and Greedy	Group Project Assigned, DP and Greedy	
12	DP and Greedy	DP and Greedy	
13	Graph Algorithms	Graph Algorithms	
14	Randomized algorithms	Midterm #3	Wednesday April 30
15	String Algorithms	String Algorithms	
16	P vs NP problem	Group Project Due, Final Exam Review	
17	Final Exam	No Class	Monday May 19, 2:45PM-4:45PM

Course Policy:

- **Attendance:** Regular attendance is crucial for understanding the course content, participating in class discussions, and achieving academic success.
- **Drops and Withdrawals:** The instructor will drop from the course any students who do not attend lectures or remain inactive until the faculty drop deadline. Inactivity includes not responding to instructor emails, not engaging with course material on Canvas, and not attending or participating in lectures. After the drop deadline, it is the student's responsibility to request a withdrawal from the course if needed.
- **Exams:**
  * **Format:** All exams in this course will be conducted in-person, on paper format, and proctored by the instructor. Electronic devices won't be allowed during exams, including calculators.
  * **Religious Observances:** If any scheduled exam dates conflict with your religious observances, please notify your instructor within the first two weeks of the semester.
  * **No Make-up Policy:** There will be no make-up exams in this course, except in cases of serious, unforeseen circumstances (e.g., hospitalization) that are beyond the student's control. If such a situation arises, students must provide documentation at least 24 hours before the exam. In the event of an excused absence, the missed exam grade will be determined by taking the grade from the final exam and dividing it by two. Please note, this policy does not apply to students who miss a midterm exam without valid justification.
- **Project:** In this project, students will work individually or collaboratively with other students to solve a real-world algorithmic problem by applying concepts learned in class. The project emphasizes step-by-step problem-solving, team collaboration, and effective communication to design, implement, and analyze algorithmic solutions. This project will deepen the student's understanding of algorithm design and analysis.
- **Practice Exercises:** these are intended for practice only and will not be collected or graded. While we will review some of the problems in class, it is the student's responsibility to complete these practice exercises independently and use them as a tool for studying and assessing their knowledge in preparation for exams.
- **Grade Appeal:** Students may request a grade review for their midterm exams within one week of the grades being posted on Canvas. To request a review, students must: (1) schedule an office hour appointment with the instructor to go over the exam, and (2) provide evidence that points were deducted in error. After this one-week period, the grade for the midterm will be final, and no further revisions will be considered.
- **Recordings:** Lectures are not recorded in this course when the modality of teaching is in-person.
- **Incomplete Grades:** The instructor does not assign incomplete grades. Students are responsible for dropping the course, requesting a withdrawal, or selecting the Credit/No Credit option if they are unable to meet the minimum requirements for the course.
- **Class Notes:** Students may not capture audio, photos or video from class sessions on their own devices without the explicit permission of the instructor and everyone present, unless part of a DPRC-authorized accommodation;
- **Office Hours Policies:** Office hours are a valuable resource provided by instructors to support students' learning and offer additional assistance outside of regular lectures. They are intended to foster a deeper understanding of course materials, answer specific questions, provide guidance on assignments, and address any concerns related to the course. It is important to emphasize that office hours are not intended to serve as a replacement for missed lectures. Instructors do not typically use office hours to re-lecture or repeat the content covered during a missed class.

- **Instructor Availability:** I am unable to respond to emails or messages on weekends or holidays. You may still send an email or message, but I will not be able to respond until the next business day.
- **Final Grades:** I create a lot of opportunities for students to do well in this course, and all posted grades are final unless there was a verifiable mistake in the grade calculation. Fairness demands, and university rules, require that all students be marked according to the same standards, so that all students receive the marks they earned. I can't just raise your final grade for unjustifiable reasons, so I will not respond to any such requests.
- **Syllabus Updates:** This syllabus and schedule are subject to change. The official syllabus will be maintained at the course website. The instructor will make an announcement every time a policy from the syllabus is updated during the semester. It the student responsibility to check on the site frequently and even when revisions are made while you are absent.
- **Academic Integrity:** Any form of cheating or plagiarism will incur very serious consequences. Students may not post any course materials to any third-party sites (such as Chegg) or post any recordings, screenshots, audio or chat transcripts in any setting outside the class, and that violations of this are subject to student disciplinary action. In addition, the use of artificial intelligence (Al) or other automated writing tools to complete assignments is strictly prohibited in this class. In addition, any evidence of the use of Al will be considered a violation of academic integrity and will be met with a failing grade for the assignment and may result in further disciplinary action. It is the responsibility of each student to ensure that all work submitted for this class is their own original work, written and completed without the use of Al or other automated writing tools. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee. Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University.
  - To learn more about the university policies regarding academic integrity and other policies:
    1. Academic Dishonestly: https://conduct.sfsu.edu/academic-integrity
    2. SFSU Code of Student Conduct: https://conduct.sfsu.edu/standards
    3. Plagiarism: https://conduct.sfsu.edu/plagiarism
    4. CS Department Student Policies: https://cs.sfsu.edu/grads/student-policies
- **Students with disabilities:** Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice 415-338-2472, video phone 415-335-7210) or by email (dprc@sfsu.edu).

  SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the Title IX Coordinator by completing the report form available at http://titleix.sfsu.edu, emailing vpsaem@sfsu.edu or calling 338-2032. To disclose any such violence confidentially, contact: The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/ Counseling and Psychological Services Center - (415) 338-2208; http://psyservs.sfsu.edu/ For more information on your rights and available resources: http://titleix.sfsu.edu

- **Religious and cultural holidays policy** The religious and cultural holidays policy (S09-212) states that the faculty of San Francisco State University shall accommodate students wishing to observe religious and cultural holidays when such observances require students to be absent from class activities. It is the responsibility of the student to inform the instructor, in writing, about such holidays during the first two weeks of the class each semester. If such holidays occur during the first two weeks of the semester, the student must notify the instructor, in writing, at least three days before the date that they will be absent. It is the responsibility of the instructor to make every reasonable effort to honor the student request without penalty, and of the student to make up the work missed.
- **Academic calendar policy:** The Academic Calendar can be found at https://webapps.sfsu.edu/public/webcal/acadcalendar